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Asia-Pacific Technology | Asia Pacific

AI Supply Chain: TSMC to expand 3nm capacity for major AI customer's growth?

Additional 3nm wafer supply would fuel more growth of AI semis in 2026. We also see potential upside to our existing TSMC capex of US\$43bn in 2026.

Major AI customer asks TSMC to build more wafer capacity. In our previous [AI Supply Chain tracker](#), we mentioned that CoWoS was not really the major constraint factor; rather it was TSMC's wafer front-end and ABF substrate supply in 1H26 (give the T-Glass shortage). Alchip also reaffirmed a 3nm wafer shortage ([link](#)). We held the view that TSMC 2026 capex could only be US\$43bn (up from US\$41bn in 2025), as TSMC wanted to keep its leading-edge capacity full, not just for 2026, but also future years. However, after Nvidia (covered by Joseph Moore) CEO's Taiwan visit last weekend, the plan for its 3nm capacity may have changed (see [Nvidia CEO Huang sees strong demand for Blackwell chips](#)).

1. Our channel checks suggest an additional 20k of 3nm capacity could be added in Taiwan fab: According to TSMC's management on its earnings call, all new clean room space in Taiwan would be for 2nm expansion, while 3nm expansion would need to happen in existing fabs. Now, since TSMC has decided to add 20kwpm more 3nm in Taiwan, clean room space is becoming the key issue. Our latest checks suggest TSMC may move 22nm/28nm out of Fab15 (and reserve those tools for a future Europe fab). By adding 20k more 3nm capacity, or US\$5-7bn capex (assuming US\$300mn per k), would imply its 2026 capex reaching US\$48-50bn, similar to bull camp assumptions – a potential positive catalyst to semicap.

2. Follow up research on our power capacity vs. CoWoS capacity. Our analysis on what 10GW of Nvidia's Rubin chips would mean to TSMC's CoWoS capacity has gained traction. By increasing the project pipeline (other mega and hyper power deployment plans), we still conclude that TSMC's CoWoS capacity should be sufficient (see [Exhibit 3](#)). Please also see [Exhibit 18](#) for our cloud capex and AI server capex update.

3. ASIC design service update: TSMC and its subsidiary GUC are working together to service Tesla (covered by Adam Jonas), and enable its 3nm AI5 design and production. Of note, AI5 will continue to use LPDDR instead of HBM, and hence we are not changing our TSMC CoWoS capacity assumptions. We raised our GUC PT to US\$2,288 ([link](#)). And in our [TSMC: Thoughts about auto semi foundry dynamics](#), we mentioned a US\$2bn foundry opportunity a year for the 2nm Tesla AI6 chip.

Stock implications: We stay OW on TSMC (Top Pick)/KYEC/ASE on potential better AI semi growth. The potential 3nm capex hike could be positive to global semicap sentiment. We are also OW on selective ASIC names, such as GUC/Alchip.

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2026 AI supply upside may hinge on TSMC 3nm capacity

Our assumption for TSMC 3nm capacity is that it should be around 110-120kwpm for 2025 and then grow to 140-150kwpm by 2026. The incremental capacity includes 20kwpm from TSMC's AZ fab phase 2 and another 10kwpm conversion from 4/5nm in Taiwan. However, we have been hearing that several TSMC customers may have had difficulty securing enough 3nm capacity at TSMC, including Nvidia, AMD and Alchip (see [Nvidia CEO Huang sees strong demand for Blackwell chips](#)).

As a result, our recent checks suggest that TSMC is now considering to expand 3nm capacity 20kwpm further than our initial expectation. However, clean room space is the problem for expansion, as TSMC currently cannot find clean room for it (all the new clean room space will be for 2nm, according to TSMC's management on the earnings call). Our latest checks suggest that TSMC may move its 22nm/28nm out from Fab15, and move to a European fab. If this took place, it would imply 2026 capex rising to US\$48-50bn, assuming US\$300mn capex per k wafer of 3nm.

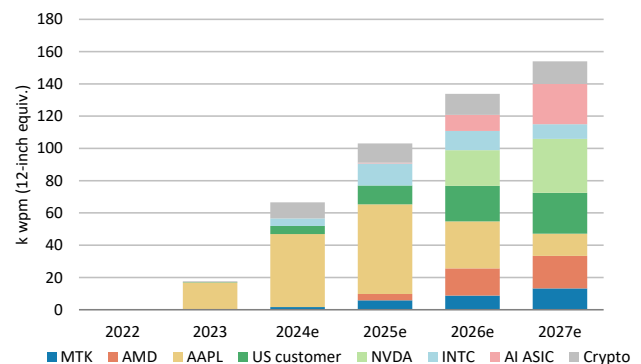
Exhibit 1: TSMC 3nm capacity expansion change

	Old MSe	New MSe
2025 capacity	110-120kwpm	110-120kwpm
2026 capacity	140-150kwpm	160-170kwpm
	20kwpm from AZ fab	20kwpm from AZ fab
Additional Capacity	10kwpm from N4/5 conversion	10kwpm from N4/5 conversion Another 20kwpm at Fab 15

Source: Morgan Stanley Research estimates

Exhibit 2: TSMC 3nm customer demand breakdown

Demand for TSMC's 3nm node



Source: Morgan Stanley Research (e) estimates

Calculating the power deployment demand for CoWoS

We published our power deployment implied CoWoS analysis in the last issue of our AI tracker ([see note](#)). Many investors were curious about what the currently announced power deployment plans implied for CoWoS demand. We thus gathered the mega and hyper power deployment plans that we see right now, and if the usage of chips is not specified, we use NVIDIA GPU and ASICs as a reference. We calculated the racks needed for each project based on power consumption, and summed up the chips volume housed in these racks. Thus, the implied CoWoS number is 1.24mn for total consumption, which implies 681k units of CoWoS demand annually vs. TSMC and non-TSMC CoWoS average annual capacity of nearly 1,100k. We think this is in-line with our understanding that CoWoS is not the key bottleneck, while front-end capacity and other materials, such as T-glass, are the potential products in shortage.

Exhibit 3: Announced mega and hyper power deployment plans, and implications for TSMC and non-TSMC CoWoS supply chain

Mega Power Deployment Project (>1 GW)													
Company	Project name and deployment site	Power Required (GW)	Status	Region	Major Purpose	Targeted On-line Date	Disclosed date	Source	Rack name	Power consumption per Rack (kW)	Rack Number (k units)	Chip Name	Implied Annual CoWoS Volume (k wafers)
OpenAI/Oracle/SoftBank	Stargate - Abilene, TX	1	Under Construction	North America	AI Training	2025-2026	Jan-25	OpenAI	GB200 NVL72	135	7	Blackwell	533 38 19
	Stargate US Expansion (5 sites)	5	Planned/signed	North America	AI Training	2026-2028	Jul-Sep 2025	OpenAI	Vera Rubin NVL144	220	23	Rubin GPU	1636 205 68
	Stargate - Multiple US Sites	1.5	Planned	North America	AI Training	2026-2027	Sep-25	Data Center Executive	Vera Rubin NVL144	220	7	Rubin CPU	491 61 31
	Stargate - Multiple US Sites	2.5	Planned	North America	AI Training	2027-2030	Jan-25		Vera Rubin NVL144	220	11	Rubin GPU	245 12 4
												Rubin CPU	818 102 26
Stock Farm Road/Fir Hills	Jeollanam-do, South Korea	3	Planned	APAC	AI/Hyperscale	2028	Feb-25	Unofficial Engineering	Vera Rubin NVL144	220	14	Rubin GPU	982 123 123
Meta	Hyperion - Louisiana	2	Under Construction	North America	AI Training	2030	Dec-24	DCO	Vera Rubin NVL144	220	9	Rubin GPU	655 82 82
		3						Santa Barbara	Vera CPU	327 16 5			
Amazon AWS	New Carlisle, Indiana	2.2	Under Construction	North America	AI/Cloud	2026-2030	Apr-24	Chalk	Teton 3	100	22	Trainium 3	1584 158 32
Microsoft	Fairwater - Wisconsin (Phase 1+2)	1.5	Under Construction	North America	AI Training	2026-2029	Mar-23	Microsoft	Vera Rubin NVL144	220	7	Rubin GPU	491 61 15
xAI	Colossus 1 - Memphis, TN	0.3	Operational/Expanding	North America	AI Training	2025-2026	Jun-24	Sentientia	Hopper HX	18	8	Hopper GPU	67 2 1
	Colossus 2 - Memphis, TN	1	Under Construction	North America	AI Training	2025-2026	Mar-25	Bloomberg	GB200 NVL72	135	7	Blackwell	533 38 19
Google	Nebraska/Iowa Cluster	1	Under Construction	North America	AI Training	2025-2026	2023-2024	Bloomberg	TPU	60	17	TPUv7 (Ironwood)	1067 17 8
Anthropic	Location TBD (SMR Nuclear)	1	Planned	North America	TBD	2026	2025		TPU	60	17	TPUv7 (Ironwood)	1067 17 17
Oracle		1							Vera Rubin NVL144			Rubin GPU	327 41 41
ECL	TerraSite-TX1 - Houston, TX	1.0 (up to 2.0)	Planned	North America	AI/Cloud	2025-2027	Oct-24	Data Center Executive	Vera Rubin NVL144	220	5	Vera CPU	164 8 3
Hyperscale (200-999 MW)													
Company	Project name and deployment site	Power Required (MW)	Status	Region	Major Purpose	Targeted On-line Date	Disclosed date	Source	Rack name	Power consumption per Rack (kW)	Rack Number (k units)	Chip Name	Implied Annual CoWoS Volume (k wafers)
Amazon AWS	New Carlisle - Phase 1	525	Operational	North America	AI/Cloud	2024	Apr-24	OpenText	Trainium 1/ Inferentia 2 chip produced				600 30 30
									Trainium 2/2.5 chip produced				350 10 10
Microsoft	Wisconsin - Phase 1	400	Under Construction	North America	AI	2026	Mar-23	DCO	Vera Rubin NVL144	220	2	Hopper GPU	280 10 10
									Rubin CPU			131 16 16	
Google	Fort Wayne, Indiana	200	Planned	North America	Cloud	TBD	2024	Data Center Executive	TPU	60	3	TPUv7 (Ironwood)	65 3 1
													213 3 2
Implied annual CoWoS demand													1,243 681

Source: Company data, Morgan Stanley Research estimates

We also try to calculate the demand for both TSMC and non-TSMC CoWoS based on the currently announced partnerships, including the OpenAI-NVIDIA 10GW, OpenAI-AMD 6GW, and Google-Anthropic 1GW contracts. Our analysis shows that total implied CoWoS consumption for TSMC stands at 629k wafers and 110k for the non-TSMC camp. By assuming that both the OpenAI-NVIDIA and OpenAI-AMD contracts will be realized in three years, we expect 2027 annual CoWoS demand from these projects would be 245k for TSMC and 37k for the non-TSMC camp. The implied 3nm wafer consumption could be 260k for Nvidia (OpenAI), 70k for Broadcom (Anthropic), and 2nm wafer consumption of 81k for AMD (OpenAI) project, based on the announced partnerships.

Exhibit 4: Announced Power Deployment Implications for TSMC

Announced Power Deployment Implication for TSMC			
GPU/ASIC Vendor	NVIDIA (Open AI)	AMD (Open AI)	Broadcom (Anthropic)
Power Deployed (GW)	10	6	1
Rack name	Vera Rubin NVL144	Helios	TPU
Power consumption per Rack (kW)	220	260	60
Rack Number (k units)	45	23	17
Chip Name	Rubin GPU	MI455 GPU	TPUV7 (Ironwood)
Chip Volume (k units)	3,273	1,662	1,067
Implied CoWoS Volume (k wafers)	409	166	53
2027 annual CoWoS demand (k wafers)	136	55	53
Implied 2/3nm wafer Volume (k wafers)	260	81	70

Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our Asia supply chain checks; note: We assume 64units of TPU chips per rack with TPUv7 per chip TDP being ~940W

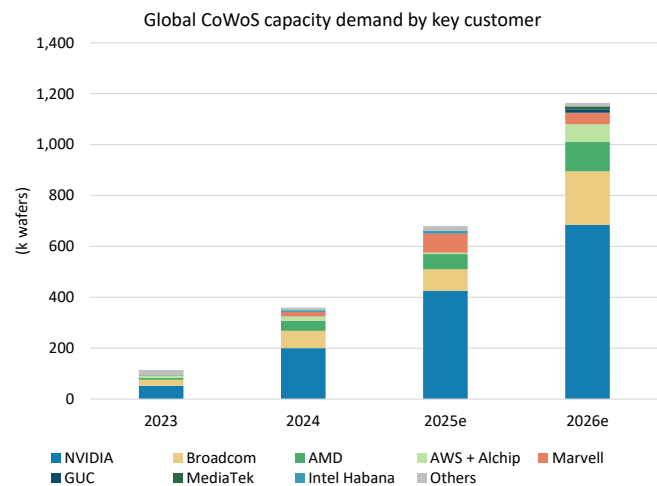
Exhibit 5: Announced Power Deployment Implications for the non-TSMC camp

Announced Power Deployment Implication for Non-TSMC Supply Chain		
GPU/ASIC Vendor	NVIDIA	AMD
Power Allocated (GW)	10	6
Rack name	Vera Rubin NVL144	Helios
Power consumption per Rack (kW)	220	260
Rack Number (k units)	45	23
Chip Name	Vera CPU	Venice CPU
Chip Volume (k units)	1,636	831
Implied CoWoS Volume (k wafers)	82	28
2027 annual CoWoS demand (k wafers)	27	9

Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our Asia supply chain checks; note: We assume 64units of TPU chips per rack with TPUv7 per chip TDP being ~940W.

AI Demand Analysis – CoWoS Capacity

Exhibit 6: Global CoWoS consumption, by customer



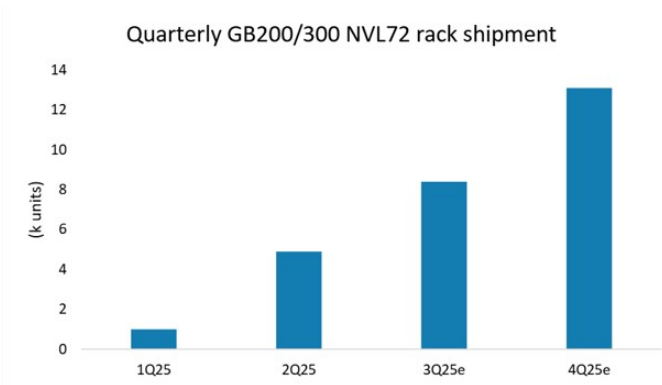
Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our Asia supply chain checks

Exhibit 8: Global CoWoS demand Y/Y growth profile

Y/Y	2023	2024e	2025e	2026e
NVIDIA	119%	280%	113%	61%
Broadcom	56%	191%	25%	147%
AMD	485%	470%	50%	92%
AWS + Alchip		71%	(69%)	1300%
Intel Habana			29%	(100%)
Marvell	(22%)	1438%	317%	(40%)
GUC		(15%)	300%	600%
Others	23%	(49%)	90%	(21%)
Total demand	95%	216%	84%	71%

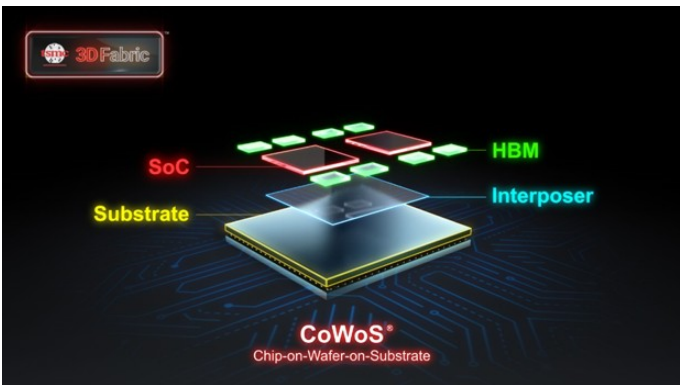
Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our Asia supply chain checks

Exhibit 7: Our downstream hardware team also estimates around 27.6k of GB200/GB300 server rack shipments in 2025e



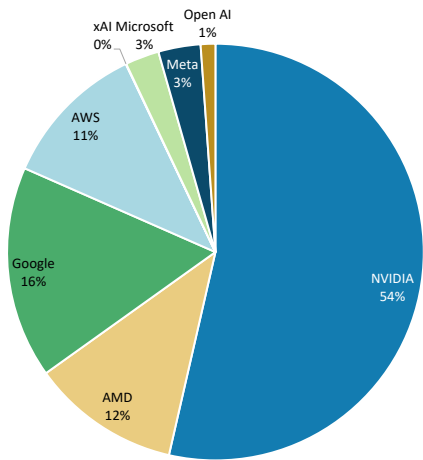
Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our Asia supply chain checks

Exhibit 9: CoWoS: Chip on Wafer (interposer) on Substrate



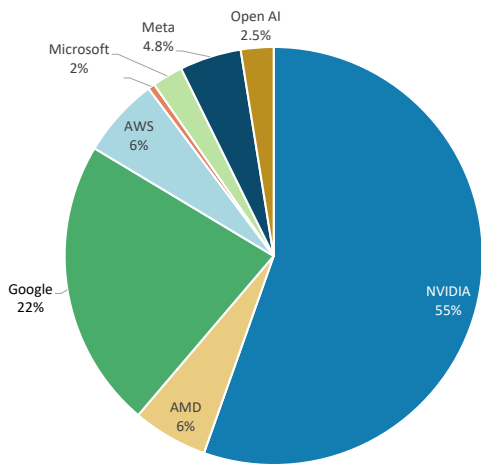
Source: TSMC

Exhibit 10: HBM consumption in 2026e breakdown by customer



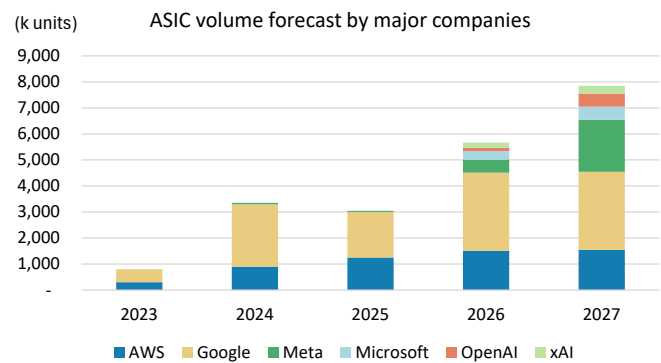
Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our Asia supply chain checks

Exhibit 11: AI computing wafer consumption in 2026e breakdown by customer



Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our Asia supply chain checks

Exhibit 12: ASIC volume forecast



Source: Morgan Stanley Research (e) estimates

Exhibit 13:

HBM consumption in 2026e – up to 26bn GB

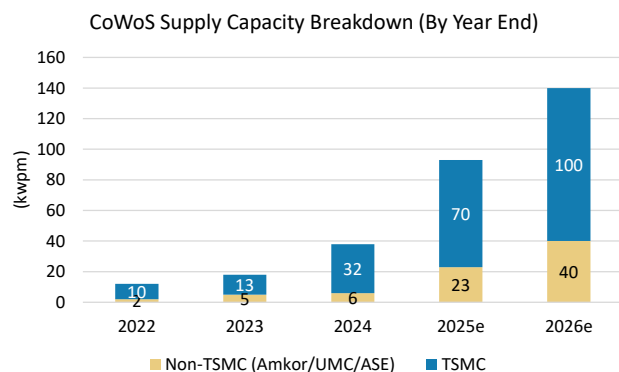
AI chip vendor	Product name	CoWoS capacity allocation (k wafers)	Chips per CoWoS wafer	Implied shipments (k)	HBM chip density (GB)	HBM chip units	Total HBM size (GB)	HBM generation	HBM vendor	Total HBM demand (k GB)
AI GPU (2026e)										
NVIDIA	B300	235	14	3,290	36	8	288	HBM3e 12hi	Hynix/Micron/Samsung	947,520
	Rubin R200	355	8	2,840	36	8	288	HBM4		817,920
AMD	MI355	10	12	120	36	8	288	HBM3e 12hi	Samsung	34,560
	MI400	80	10	800	36	12	432	HBM4	Samsung	345,600
AI ASIC (2026e)										
Google	TPU v6 (Trillium)	20	20	400	24	4	96	HBM3e 8hi	Samsung/Micron	38,400
	TPU v7 (Ironwood)	120	20	2,400	24	8	192	HBM3e 8hi	Hynix/Samsung	460,800
	TPU v8 (Zebrafish)	10	20	200	36	6	216	HBM3e 12hi	Hynix/Micron	43,200
AWS	Trainium 2.5	20	16	320	24	4	96	HBM3e 8hi	Hynix/Samsung/Micron	30,720
	Trainium 3	70	17	1,190	36	8	288	HBM3e 12hi	Hynix/Samsung/Micron	342,720
xAI		10	20	200	24	6	144	HBM3e 8hi	Samsung?	1,152
Microsoft	Maia 200	4	29	116	16	4	64	HBM3	Samsung	7,424
	Maia 300	25	11	275	36	8	288	HBM4	Samsung?	79,200
Meta	MTIA 3 (Iris)	50	10	500	36	6	216	HBM3e 12hi	Hynix/Samsung	108,000
Open AI	Nexus (Titan 1)	10	13	130	36	6	216	HBM4	Hynix	28,080
Total		1,109								3,285,296
Total HBM demand (mn Gb)										26,282

Source: Company data, Morgan Stanley Research (e) estimate. Note: Estimates are compiled using our Asia supply chain checks

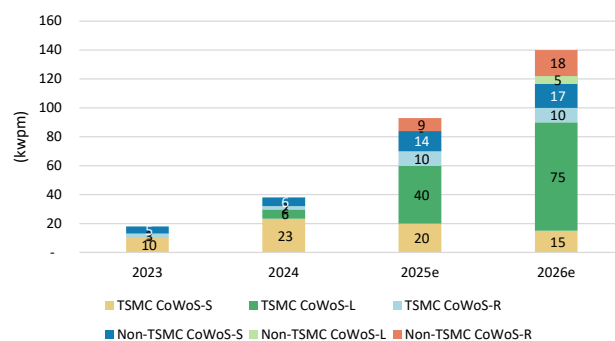
Exhibit 14: AI computing wafer consumption in 2026e – up to US\$21bn revenue

AI chip vendor	Product name	CoWoS capacity allocation (k wafers)	Chips per CoWoS wafer	Implied shipments (k)	Compute die size	Geometry	Compute die units	Wafer consumption (k wafers)	Wafer price (US\$)	Wafer revenue TAM (US\$ mn)
AI GPU (2026e)										
NVIDIA	B300	235	14	3,290	850	4nm	2	261	21,945	5,730
	Rubin R200	355	8	2,840	850	3nm	2	225	24,300	5,477
AMD	MI355	10	12	120	110	3nm	8	3	24,300	85
	MI400	80	10	800	110	2nm	8	39	28,125	1,091
AI ASIC (2026e)										
Google	TPU v6 (Trillium)	20	20	400	600	5nm	2	18	19,609	352
	TPU v7 (Ironwood)	120	20	2,400	700	3nm	2	157	24,300	3,812
	TPU v8 (Zebrafish)	10	20	200	800	3nm	2	15	24,300	363
AWS	Trainium 2.5	20	16	320	600	5nm	2	9	20,543	187
	Trainium 3	70	17	1,190	700	3nm	2	41	26,089	1,061
xAI		10	20	200	645	4nm	1	5	21,945	107
Microsoft	Maia 200	4	29	116	700	3nm	1	3.0	24,300	74
	Maia 300	25	11	275	850	2nm	1	14.6	28,125	409
Meta	MTIA 3 (Iris)	50	10	500	850	3nm	2	39.7	24,300	964
Open AI	Nexus (Titan 1)	10	13	130	750	3nm	4	18.2	28,125	512
Total		1,109						849		20,224

Source: Company data, Morgan Stanley Research (e) estimate. Note: Estimates are compiled using our Asia supply chain checks

Exhibit 15: TSMC and non-TSMC CoWoS capacity

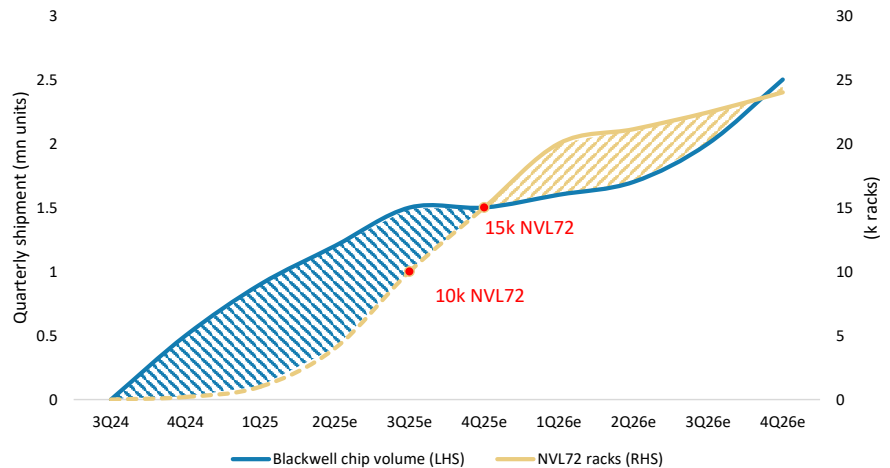
Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our Asia supply chain checks

Exhibit 16: Detailed CoWoS capacity expansion by year end and by vendor

Source: Company data, Morgan Stanley Research (e) estimates; note: estimates are compiled using our Asia supply chain checks

Global AI capex: remains strong after Oracle's print

Exhibit 17: TSMC is expected to produce 5.1mn chips in 2025, which could be fully digested in 1H26 based on Oracle's announced strong AI server build in 2026



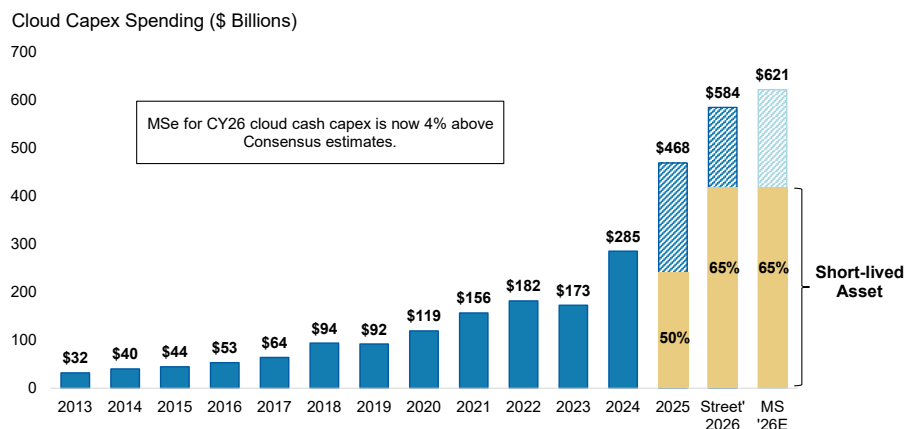
Source: Company data, Morgan Stanley Research (e) estimates. Note: Estimates are compiled using our supply chain checks and may differ from other Morgan Stanley Research estimates

Hyperscalers' capex remains strong into 2026

According to our US hardware analyst Erik Woodring's Global Cloud Capex Tracker, Morgan Stanley expects 2026 cloud capex to increase to US\$621bn, implying 33% Y/Y growth (vs. consensus now at only +25% Y/Y).

Assuming the proportion of AI server capex (spending on short-lived assets) increases, implied AI server capex may grow ~73% Y/Y in 2026 (see [Exhibit 18](#)). This underscores our Attractive industry view on Greater China Semis – we see AI demand remaining strong. In AI semis, and we are OW on TSMC, Samsung Electronics, Alchip, Aspeed, and KYEC.

Exhibit 18: By assuming an increase in the mix for AI server capex, the implied AI server capex should grow 73% Y/Y in 2026, based on Morgan Stanley estimates

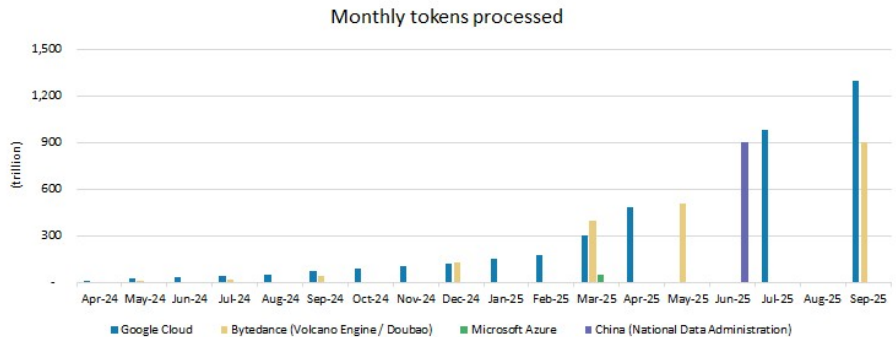


Source: Company data, FactSet, Morgan Stanley Research estimates (short-lived assets are MSe). Note: Cloud capex includes announced capex from Alphabet, Amazon, Microsoft, Meta Platforms, Alibaba, Tencent, Baidu, CoreWeave, Apple, IBM, and Oracle. ORCL forecasts are not updated post latest earnings. Forward estimates are consensus est. for the others.

Monthly tokens processed by major CSPs suggest that AI inference demand is growing

We see growing AI inference demand, as evidenced by the rapidly growing number of tokens processed by major CSPs. China's token consumption reached 30trn daily as of end-June 2025 (monthly run-rate was 900trn), +300x vs. 0.1trn daily at the beginning of 2024 (as reported by the National Data Administration of China). Google (covered by Brian Nowak) processed over 1,300trn tokens in September 2025, +33% from 980trn in July 2025. Bytedance's token consumption reached 30trn daily as of end-September 2025 (monthly run-rate was 900trn), +83% vs. 16.4trn daily as of end-May 2025. Microsoft (covered by Keith Weiss) processed over 500trn tokens through its Foundry APIs in F2025 (year end June 2025), up over 7x Y/Y.

Exhibit 19: Monthly tokens processed

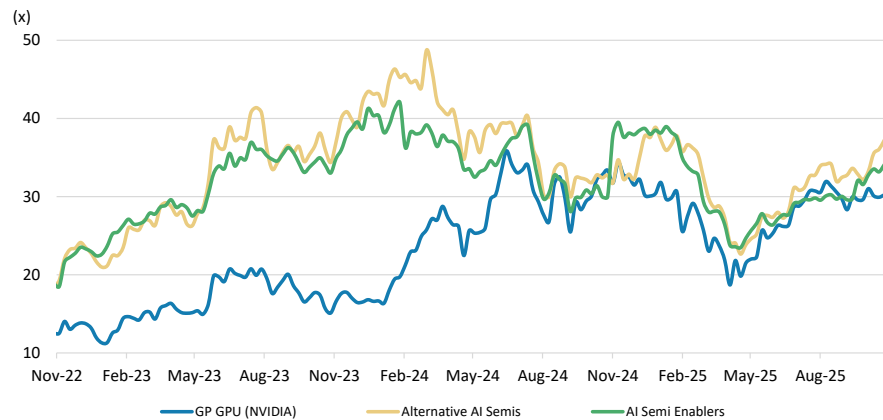


Source: Company information, Morgan Stanley Research. Note: Bytedance and China numbers represent monthly token use run-rate based on end-of-month daily tokens

AI semis – stock implications, P/E multiples, revenue exposure

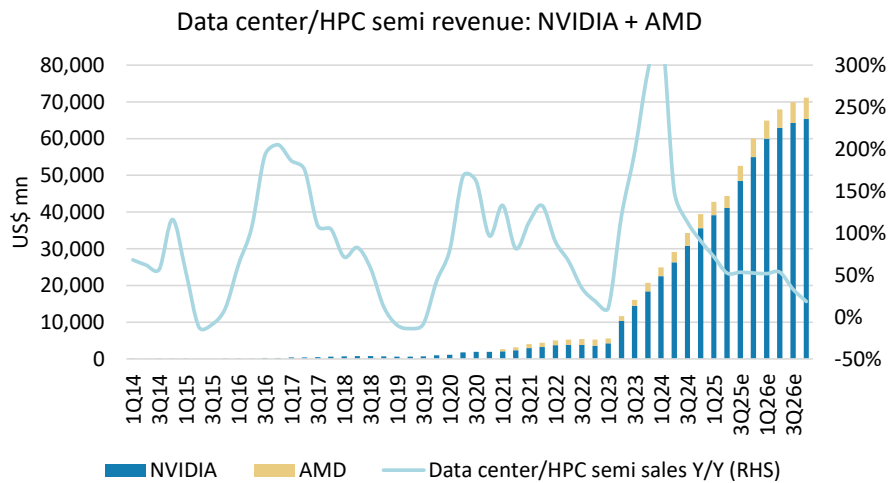
- In Asian semis, we are OW TSMC, KYEC, ASE, Samsung and Aspeed.
- We are also positive on Asian ASIC design service providers Alchip and GUC, and CPO suppliers FOCI and Himax.

Exhibit 20: P/E multiple trend of AI semis

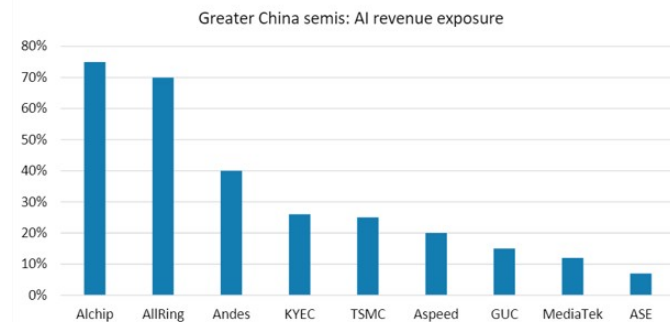


Alternative AI semis group: AMD, Alchip, Andes, Marvell, Broadcom. AI semi enablers group: TSMC, Synopsys, Cadence, ASML, BES, Ibiden, KYEC, Advantest. Source: Company data, Morgan Stanley Research.

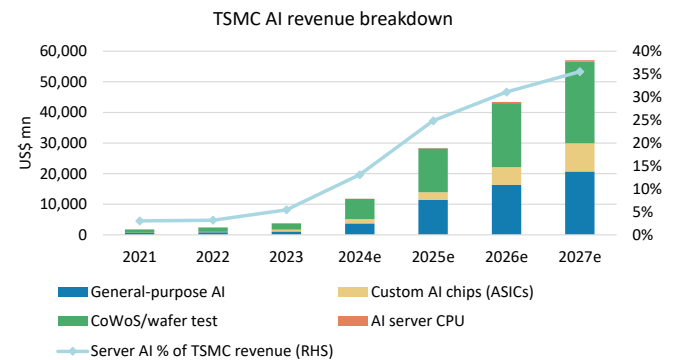
Exhibit 21: We continue to see AI chip quarterly revenue increasing



Source: Company data, Refinitiv, Morgan Stanley US Research (e) estimates.

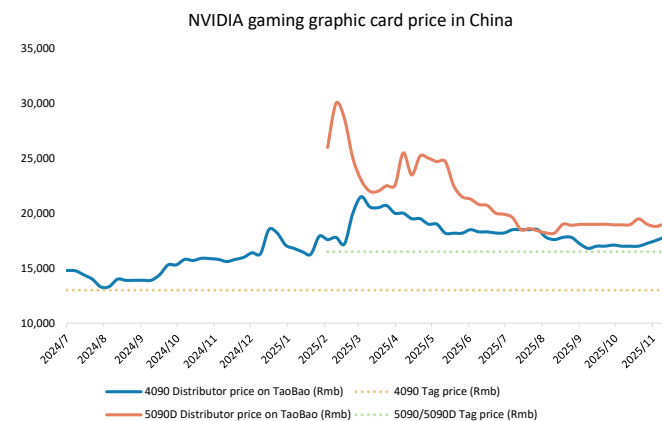
Exhibit 22: Greater China AI semi revenue exposure as of 2025e

Source: Morgan Stanley Research (e) estimates

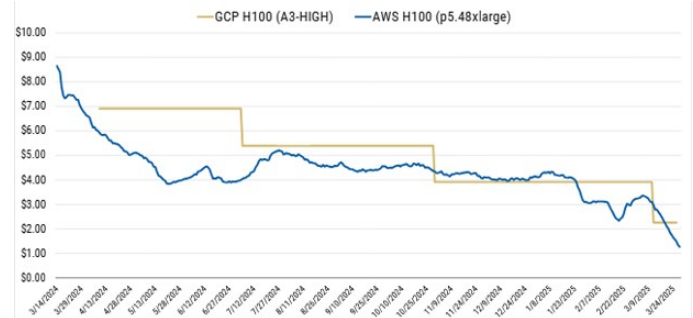
Exhibit 23: AI accounts for a mid-teens share of TSMC's 2024 total revenue and 25% of TSMC's 2025e total revenue

Source: Company data, Morgan Stanley Research (e) estimates

AI GPU and ASIC rental price tracker

Exhibit 24: Retail price of NVIDIA 4090 and 5090 graphic cards stabilized, although China's AI inference demand is still very strong

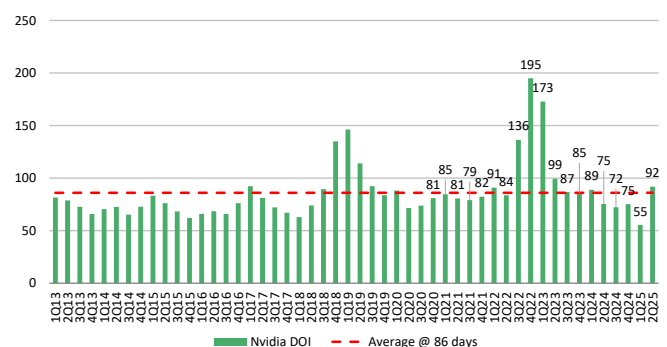
Source: TaoBao, Morgan Stanley Research

Exhibit 25: AI GPU H100 per GPU per hour as of end-March

Source: Company data, Morgan Stanley Research

Exhibit 26: AI ASIC equivalent computing power – 16x Inferentia 2 per hour

Source: Company data, Morgan Stanley Research

Exhibit 27: NVIDIA's inventory days/level

Source: Company data, Refinitiv, Morgan Stanley Research

Key featured reports on the AI supply chain

Asia-Pacific Technology: AI Supply Chain: Reconciling the strong AI infrastructure plan with TSMC's prudent CoWoS expansion (29 Oct 2025)

Asia-Pacific Technology: AI Supply Chain: TSMC outlook, ASIC updates, OCP implication for AI Semis (20 Oct 2025)

Asia-Pacific Technology: AI Supply Chain: TSMC CoWoS, Meta ASIC, and China GPU (8 Oct 2025)

Asia-Pacific Technology: AI Supply Chain: Oracle capex, OpenAI ASIC, SEMICON Taiwan (16 Sep 2025)

AllRing Tech Co.: Fundamentals Bottoming Out with 2026 CoWoS Capacity Showing Upside; Lift to OW (23 Sep 2025)

Global Technology: AI Supply Chain: AI capex revised up, while TSMC 2026 CoWoS supply unchanged (5 Aug 2025)

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Global Technology: Supply-chain Reorientation (24 Jul 2025)

Global Technology: China – AI: The Sleeping Giant Awakens (13 May 2025)

Global Technology: AI Cloud Capex in the Spotlight (26 Feb 2025)

Global Semiconductors: AI ASIC 2.0: Potential winners (15 Dec 2024)

Global Technology: Global Technology – Dawn of the AI Smartphone Era: Edge AI – Apple Intelligence Fuels Innovation – More Charts, Fewer Words (17 Jul 2024)

Global Technology: AI PCs To Usher In The Next Leg Of PC Market Growth (21 May, 2024)

Global Semiconductors: Windows on Arm AI PC – This Time is for Real (7 May 2024)

Key Upstream AI supply chain companies

Global Unichip Corp: Substantial 2027 revenue from US projects; OW (11 Nov 2025)

ASMPT Ltd: First take: Stronger-than-expected 4Q guidance; OW (29 Oct 2025)

MediaTek: Rising OSAT and Memory Costs Create More Margin Headwinds into 2026; EW (28 Oct 2025)

Egis Technology Inc: Egis Tech Day – Key Takeaways (28 Oct 2025)

Greater China Semiconductors: Backend Observations: Taiwan OSAT to Hike Price in 2026 (28 Oct 2025)

JCET Group Co Ltd: 3Q25 earnings miss; UW (23 Oct 2025)

Greater China Semiconductors: China Foundry: Upgrade SMIC to OW on Strong Domestic AI GPU Demand (21 Oct 2025)

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	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
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Equal-weight/Hold	1609	44%	370	44%	23%	788	46%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	568	15%	80	10%	14%	223	13%
Total	3,682		835			1715	

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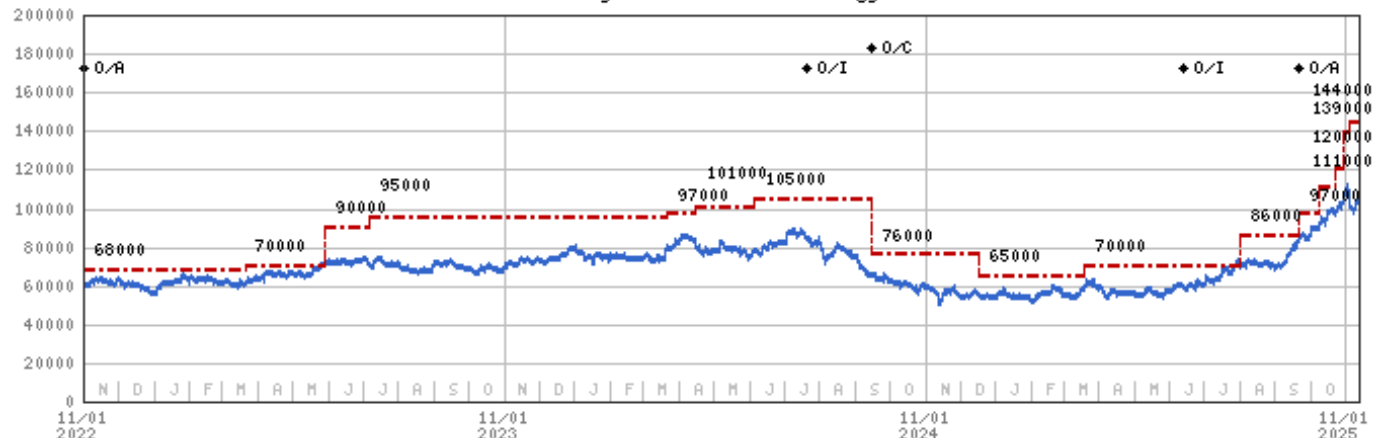
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Samsung Electronics (005930.KS) - As of 11/12/25 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 11/1/20 : O/A; 7/19/21 : O/I; 8/12/21 : O/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : O/C; 6/13/25 : O/I; 9/21/25 : O/A

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INDUSTRY COVERAGE: S. Korea Technology

COMPANY (TICKER)	RATING (AS OF)	PRICE* (11/12/2025)
Ryan Kim		
Ecopro BM (247540.KQ)	U (03/20/2023)	W155,300
Fadu Inc (440110.KQ)	O (09/21/2025)	W26,750
Hanmi Semiconductor Co. Ltd. (042700.KS)	O (08/16/2024)	W131,100
HD Hyundai Electric Co Ltd (267260.KS)	O (03/25/2025)	W848,000
Isu Petasys Co. Ltd. (007660.KS)	O (02/03/2025)	W119,600
L&F Co Ltd (066970.KS)	E (04/03/2025)	W130,700
Leeno Industrial Inc. (058470.KQ)	O (04/03/2025)	W56,900
Lotte Energy Materials Corp (020150.KS)	U (04/03/2025)	W40,200
LS Electric (010120.KS)	E (03/25/2025)	W489,000
POSCO FUTURE M (003670.KS)	U (04/03/2025)	W215,000
SK IE Technology (361610.KS)	U (04/03/2025)	W31,800
Wonik IPS Co Ltd (240810.KQ)	O (08/08/2025)	W70,100
Shawn Kim		
LG Display (034220.KS)	E (06/11/2025)	W13,600
LG Electronics (066570.KS)	E (04/07/2025)	W92,900
LG Innotek (011070.KS)	E (03/12/2025)	W252,000
Samsung Electro-Mechanics (009150.KS)	O (07/31/2025)	W226,500
Samsung Electronics (005935.KS)	O (11/18/2019)	W78,600
Samsung Electronics (005930.KS)	O (11/18/2019)	W103,100
Samsung SDI (006400.KS)	E (12/18/2024)	W321,500
Seoul Semiconductor (046890.KQ)	U (04/04/2018)	W6,290
SK hynix (000660.KS)	O (09/21/2025)	W617,000

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.

INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (11/12/2025)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$32.64
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb302.75
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$115.50
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$3,520.00
Andes Technology Corp (6533.TW)	O (08/04/2022)	NT\$256.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$227.00
Global Unichip Corp (3443.TW)	O (07/27/2024)	NT\$1,885.00
GlobalWafers Co Ltd (6488.TWO)	O (09/19/2025)	NT\$390.00
Gudeng Precision (3680.TWO)	E (07/16/2025)	NT\$332.50
Hua Hong Semiconductor Ltd (1347.HK)	U (10/21/2025)	HK\$74.45
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$210.50
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb72.06
MediaTek (2454.TW)	E (09/23/2025)	NT\$1,245.00
Nanya Technology Corp. (2408.TW)	O (09/18/2025)	NT\$163.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb413.15
OmniVision Integrated Circuits Group Inc (603501.SS)	O (05/09/2025)	Rmb124.09
Phison Electronics Corp (8299.TWO)	O (03/11/2025)	NT\$1,315.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb71.20
Silergy Corp. (6415.TW)	O (02/13/2025)	NT\$211.50
SMIC (0981.HK)	O (10/21/2025)	HK\$73.25
TSMC (2330.TW)	O (02/07/2022)	NT\$1,475.00
UMC (2303.TW)	E (10/28/2024)	NT\$45.00
Vanguard International Semiconductor (5347.TWO)	U (08/05/2024)	NT\$92.50
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$123.00
Daisy Dai, CFA		
ASMPT Ltd (0522.HK)	O (07/24/2025)	HK\$79.30
China Resources Microelectronics Limited (688396.SS)	E (06/19/2025)	Rmb49.53
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$130.00
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb111.18
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb28.86
Innoscence (2577.HK)	E (10/13/2025)	HK\$73.30
JCET Group Co Ltd (600584.SS)	U (09/25/2024)	Rmb37.69
Shanghai Anlogic Infotech Co Ltd (688107.SS)	E (05/14/2024)	Rmb26.54
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$40.92
SICC Co Ltd (688234.SS)	E (09/03/2025)	Rmb72.30
StarPower Semiconductor Ltd (603290.SS)	O (03/01/2022)	Rmb97.13
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb76.02
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb22.56
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb65.60
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$415.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,315.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$5,760.00
Egis Technology Inc (6462.TWO)	O (10/21/2025)	NT\$129.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb156.68
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb209.50
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$40.65
Montage Technology Co Ltd (688008.SS)	++	Rmb124.68
Novatek (3034.TW)	E (11/10/2025)	NT\$390.50
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$54.30
Parade Technologies Ltd (4966.TWO)	O (10/03/2025)	NT\$652.00

Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$36.50
Realtek Semiconductor (2379.TW)	O (06/19/2025)	NT\$496.50
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb80.08
Sino Wealth Electronic (300327.SZ)	U (03/31/2025)	Rmb26.95
Winbond Electronics Corp (2344.TW)	O (03/11/2025)	NT\$64.80
WPG Holdings (3702.TW)	U (11/10/2025)	NT\$64.20
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb90.14
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	O (03/11/2025)	Rmb310.00
Tiffany Yeh		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$335.50
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$357.50
Himax Technologies Inc (HIMX.O)	O (05/09/2025)	US\$7.73
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$93.21

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